

Foundations for programming languages 3: Answers for exercises

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Exercise

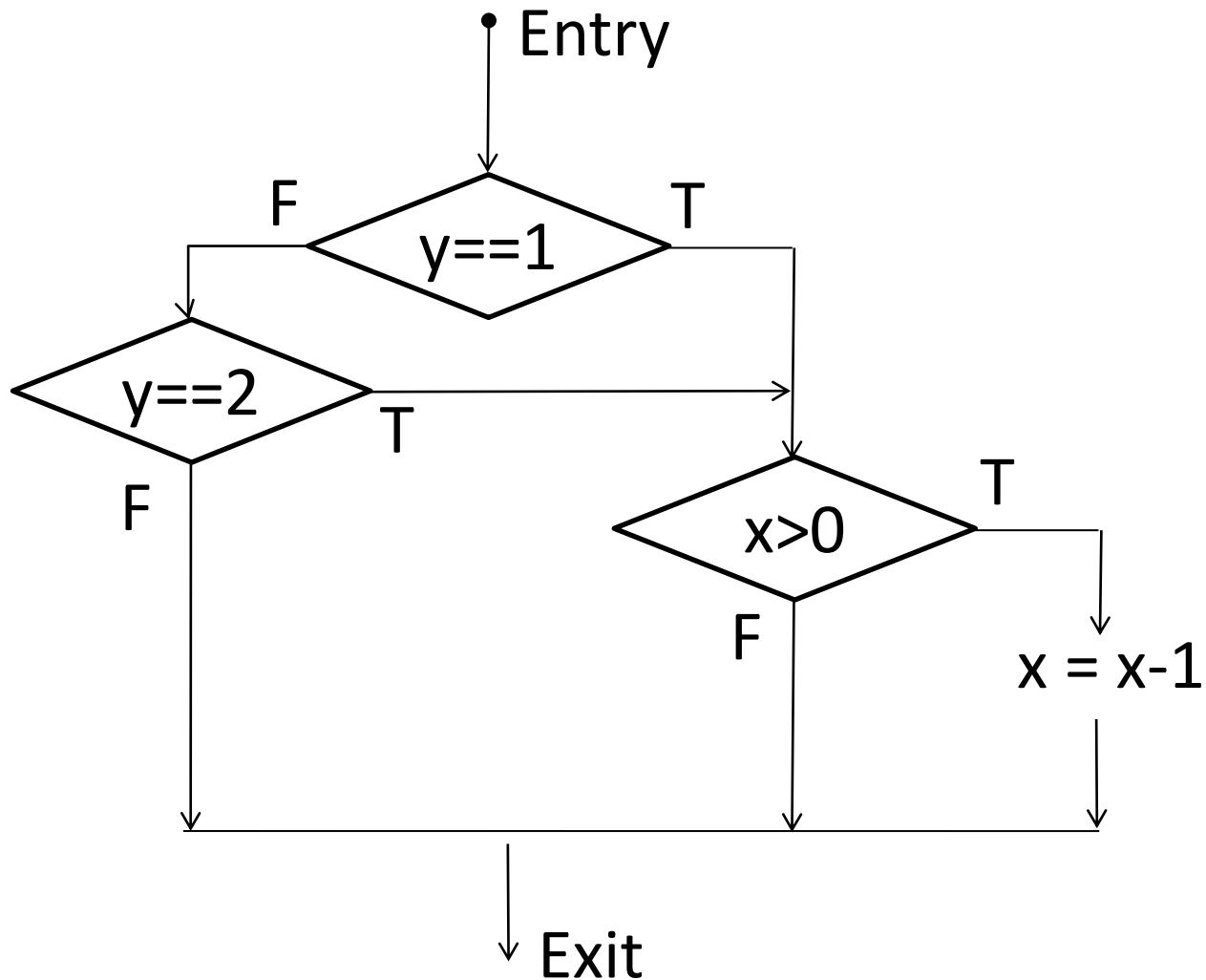
(1) Illustrate the control flow graph of the following program fragment in C.

```
if (y==1 || y==2)
    if (x > 0)
        x = x - 1;
```

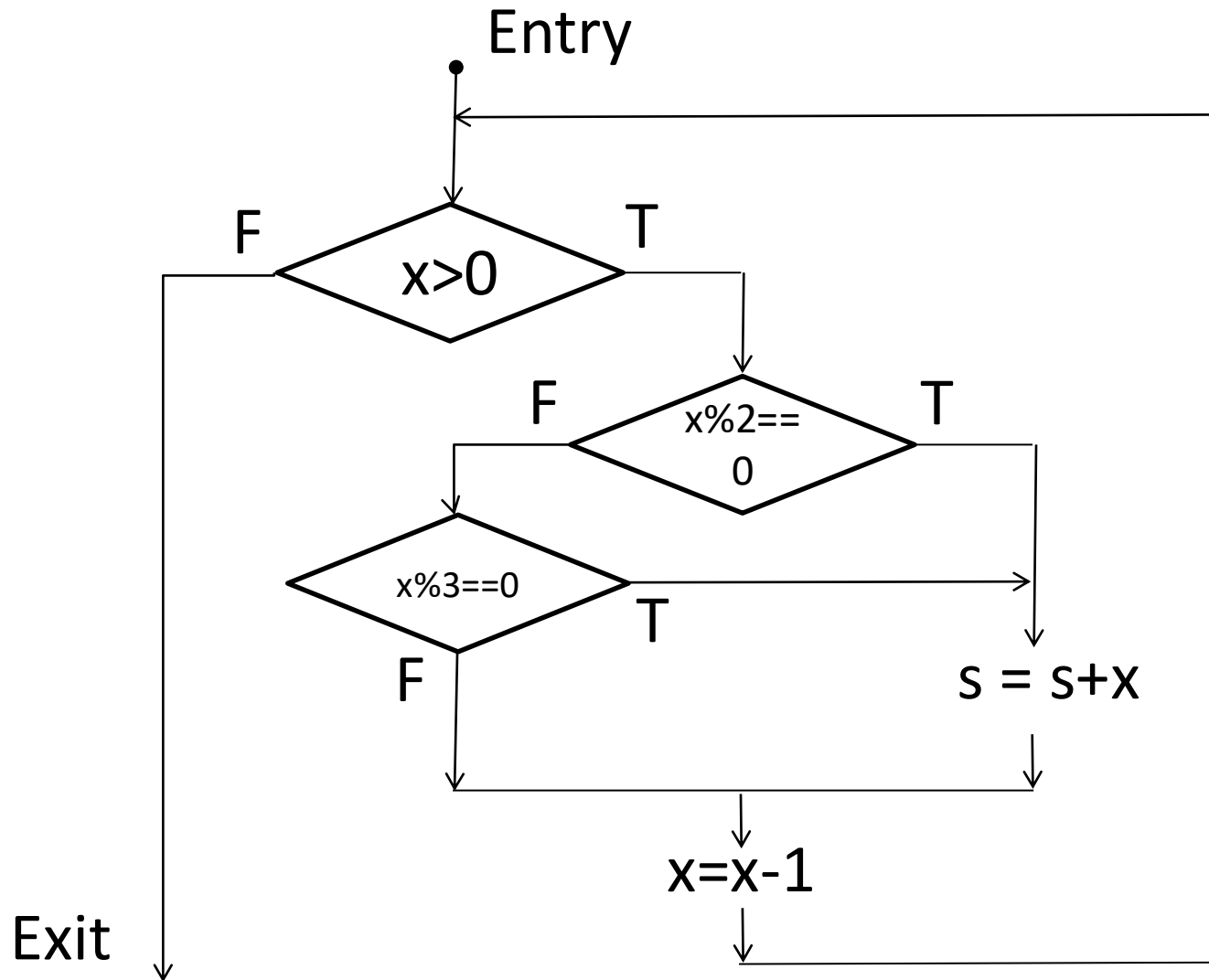
(2) Illustrate the control flow graph of the following program fragment in C.

```
while (x > 0) {
    if (x%2==0 || x%3==0)
        s = s + x;
    x = x - 1;
}
```

(1) An answer



(2) An answer



Exercises

Derive (prove) the following Hoare triples.

(1) $\{ a = 0 \} \ a := a + 2 \ \{ a = 2 \}$

(2) $\{ a = 3 \} \textbf{if } a = 3 \textbf{ then } a := a + 1 \textbf{ else } a := a - 1 \ \{ a = 4 \}$

(3) $\{ a = 1 \} \textbf{while } a < 5 \textbf{ do } a := a + 1 \ \{ a = 5 \}$

(1) An answer

$$\frac{a=0 \text{ ? } a+2=2 \quad \frac{\quad}{\{ a + 2 = 2 \} \ a := a + 2 \ \{ a = 2 \}} \text{ (assign)}}{\{ a = 0 \} \ a := a + 2 \ \{ a = 2 \}} \text{ (conseq)}$$

We abbreviate (assignment axiom) as (assign) and (consequence rule) as (conseq).

(2) An answer

See the next page.

See the next next page.

$$\frac{\frac{\{a = 3 \wedge a = 3\} \ a := a + 1 \ \{a = 4\}}{\text{See the next page.}} \quad \frac{\{a = 3 \wedge \neg a = 3\} \ a := a - 1 \ \{a = 4\}}{\text{See the next next page.}}}{\{a = 3\} \ \mathbf{if} \ a = 3 \ \mathbf{then} \ a := a + 1 \ \mathbf{else} \ a := a - 1 \ \{a = 4\}} \quad (\text{if})$$

We abbreviate (conditional rule) as (if).

(2) Cont.

$$\frac{(a = 3 \wedge a = 3) \Rightarrow a+1 = 4 \quad \{a+1 = 4\} \ a := a + 1 \ \{a = 4\} \quad a=4 \Rightarrow a=4}{\{a = 3 \wedge a = 3\} \ a := a + 1 \ \{a = 4\}}$$

(assignment)

(consequence)

(2) Cont.

$$\begin{array}{c}
 \text{(assignment)} \\
 \swarrow \\
 \hline
 (a = 3 \wedge \neg a = 3) \Rightarrow a-1 = 4 \quad \{a-1=4\} \ a := a - 1 \ \{a=4\} \quad a=4 \Rightarrow a=4 \\
 \hline
 \{ a = 3 \wedge \neg a = 3 \} \ a := a - 1 \ \{ a = 4 \} \\
 \uparrow \\
 \text{(consequence)}
 \end{array}$$

(Note)

The logical expression $(a = 3 \wedge \neg a = 3) \Rightarrow a-1 = 4$ is true because $a = 3 \wedge \neg a = 3$ is false any value of a . By the definition of $\Rightarrow$, the whole expression is true since the left side of $\Rightarrow$ is false.

(3) An answer

$$\begin{array}{c}
 \text{(assignment)} \\
 \hline
 (a \leq 5 \wedge a < 5) \Rightarrow a+1 \leq 5 \quad \{a+1 \leq 5\} \ a := a+1 \ \{a \leq 5\} \quad a \leq 5 \Rightarrow a \leq 5 \\
 \hline
 \text{(consequence)} \\
 \{a \leq 5 \wedge a < 5\} \ a := a + 1 \ \{a \leq 5\} \\
 \hline
 \text{(while)} \\
 a=1 \Rightarrow a \leq 5 \quad \{a \leq 5\} \ \mathbf{while} \ a < 5 \ \mathbf{do} \ a := a + 1 \ \{a \leq 5 \wedge \neg a < 5\} \quad (a \leq 5 \wedge \neg a < 5) \Rightarrow a=5 \\
 \hline
 \{a = 1\} \ \mathbf{while} \ a < 5 \ \mathbf{do} \ a := a + 1 \ \{a = 5\} \\
 \text{(consequence)}
 \end{array}$$